

On the relation between Voice and Case: An Agree-based approach*

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Abstract

In this article, we focus on the morphosyntactic and semantic properties of constructions involving the well-known syncretic non-active and/or reflexive morphology (depending on the language). We show that if we treat this special morphology as a property of Voice, non-active / reflexive constructions fit into a consistent, independently attested pattern of cross-linguistic variation in properties of functional heads, allowing us to derive the syntactically different constructions (reflexive, anticausative, middle, passive) exhibiting non-active/reflexive morphology in a unified manner. We argue for the general cross-linguistic schema [_{VoiceP} Voice ... [_{VP} V IA]], with Voice licensing (via Agree) the Internal Argument (IA) or the PP in all cases but in differing ways. We thus account for this seemingly disparate set of phenomena using pre-existing theoretical mechanisms, primarily Agree, in that the intrinsic interpretable features of Voice license the characteristic interpretation of the logical subject in each construction; we further contend that Voice has uninterpretable ϕ -features which probe for the interpretable ϕ -features of either the internal argument or a PP, whichever is closer in terms of c-command. We propose that non-active/reflexive constructions are distinguished by

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instantiating distinct abstract oblique Cases realised, as is standard, on the Goal of the Agree relation. Relating Case and Voice in this way takes us some way to understanding why distinct oblique Cases exist.

1. Introduction

This article focuses on the morphosyntactic and semantic properties of constructions involving the well-known syncretic non-active and/or reflexive morphology (depending on the language, as we discuss below). We show that if we treat this special morphology as a property of the syntactic category Voice, these constructions fit into a consistent, independently attested pattern of cross-linguistic variation in properties of functional heads, allowing us to derive the syntactically different constructions (i.e., reflexive, anticausative, middle, passive) exhibiting non-active or reflexive morphology in a unified manner. We argue for the general cross-linguistic schema in (1), with Voice licensing (via Agree, as we explicate below) the Internal Argument (IA) or the PP in all cases but in differing ways.

$$(1) [\text{VoiceP Voice} \dots (\text{PP}) \dots [\text{VP V IA}]]$$

While the general idea here goes back to insights in Burzio (1981, 1986), Marantz (1981, 1984), Grimshaw (1982) among others, following proposals by Labelle (2008), Ahn (2015, 2016) and others but implementing these ideas differently, our first novel claim is that in the case of reflexives, there is a Voice-feature (REFL) which marks the predicate as reflexive in the sense that reflexivity needs to be licensed by Voice, with the dimensions of variation involving (a) the exponence of Voice (i.e., the Probe) as e.g. clitic, affix, auxiliary choice, zero; and (b) the nature of the IA (i.e., the Goal) as e.g. *pro*, a weak pronoun, a special/body-part pronoun (e.g.,

X-self), which in turn are part of a more general pattern of variation, analogous to Sportiche's (1996) analysis of clitic constructions, a point we will return to in detail in section 3.

Turning to anticausatives, passives and middles, which likewise exhibit syncretic non-active morphology, we argue that the relevant non-active features are systematically associated with Voice across all these systems in the configuration in (1), with differing morphological exponence in different languages (on the connection of anticausatives and middles to Voice, see e.g. Alexiadou and Anagnostopoulou 2004 and Kastner 2016). Of crucial importance here is also the difference with respect to licensing of different PPs:

- (2) a. [VoiceP [Voice REFL] ... [VP V IA-self/body(part)]]
- b. [VoiceP [Voice A-CAUS] ... *from*P ... [VP V (IA)]]
- c. [VoiceP [Voice MIDDLE] ... [AdvP ADV *for*P] ... [VP V (IA)]]
- d. [VoiceP [Voice PASSIVE] ... *by*P ... [VP V (IA)]]

The optional PPs here correspond to the external argument; whether the PP actually instantiates the external argument is a point we take up in section 8:

- (3) a. The window broke from the pressure.
- b. This book reads easily for children.
- c. John was arrested by the police.

To (2) we can add Active Voice:

- (4) [VoiceP [Voice ACTIVE] [VP [VP V IA]]

Here Voice licenses the IA as accusative.

The real question of course is what underlies these alternations. We propose that (2b,c,d) are distinguished by instantiating distinct abstract oblique Cases: ablative (*from*), dative (*for*), instrumental/ergative (*by*). The Cases are realised on the Goal of the Agree relation as is standard, see Chomsky (2000). Relating Case and Voice in this way takes us some way to understanding why distinct oblique Cases exist. In some languages (e.g. Albanian, Greek, Latin), there is syncretic realisation of ablative and dative. To this we can add that Active Voice licenses an Accusative IA. If Voice can license exactly one Case, then we understand why the IA cannot be accusative in (2b,c,d) and must therefore move. If REFL licenses *-self*, then reflexive marking can also be seen as Case (Pesetsky 2011).

A more formal account of the Agree relation between Voice and the IA in the respective constructions would run as follows: the intrinsic interpretable features of Voice license the characteristic interpretation of the logical subject in each construction, but we assume that Voice has uninterpretable ϕ -features which probe for the interpretable ϕ -features of the closest possible c-commanded Goal: PP in (2b,c,d), X-self in (2a), IA in (4). We thus conjecture that the oblique Cases are instantiations of these (interpretable) ϕ -features. In these terms, oblique Cases represent interpretable features, whereas structural Cases represent uninterpretable features. Formulating Agree in this way in these constructions leads us to see a tight relation between Voice, Case and (ϕ -)Agree. The fact that, as we will see below, many languages have a single non-active marker for (2b,c,d) clearly supports this approach, although participial passives are a systematic exception for reasons which we will speculate on in section 8.

In what follows, we will unpack the statements above. We begin in section 2 by offering a systematic comparison of Voice and AgrS, by which we mean the agreement features associated with subjects in finite clauses whether they are instantiated on a separate head or are

features of Tense.¹ We will see that the two functional heads have very similar properties in terms of their intrinsic features, their morphosyntactic exponents and their ability to license arguments. The principal differences between them largely follow from the fact that AgrS is located very high in the inflectional field, while Voice occupies a low position, probably the lowest position in the inflectional field. In this way, our theory of Voice is fully integrated into the general theory of agreement and clause structure.

The article is organised as follows. In section 2, we show the parallels between Voice and subject agreement just alluded to. Section 3 looks at the different form of the morphological exponents of Voice and introduces Sportiche’s (1996) analysis of “Clitic Voices” in detail. In the remainder of the article, we will look at each construction type in turn -- i.e., reflexives (including deponents) in section 4, anticausatives in section 5, middles in section 6, and passives in section 7 -- arguing that non-active features are systematically associated with Voice in all cases, with differing morphological exponence, as we summarise in section 8. In section 9, we suggest a tentative extension of our analysis to the superficially very different voice systems in Austronesian and Mandarin, concluding in section 10.

¹ It is worth noting that the argument against Agreement projections (AgrSP and AgrOP) in Chomsky (1995: 321-326) relies on the idea that checked uninterpretable features are deleted, with the problematic result that the Agreement projections must disappear at the C-I interface. However, if we assume a different approach to “feature checking”, for example one based on a notion of “dependent feature” as in Zeijlstra (2020) or, in a similar vein, Form Copy (Chomsky et al. 2023, Roberts 2024, Kallulli and Roberts 2025a), then this conclusion does not follow. Furthermore, well-known facts concerning past participle agreement in many Romance languages show that checked features do not always delete, an argument first put forward in Marantz (1995). Consider for example (i):

- (i) *Le* *mele* *sono* *state* *mangiate*.
 the_{FEM,PL} apples_{FEM,PL} be_{3PL} been_{FEM,PL} eaten_{FEM,PL}
 ‘The apples have been eaten.’

Here, the plural feature is morphologically instantiated five times over, in fact in every word; the *-e* ending on the definite article, the plural form of ‘apple’ and both participles marks feminine gender and plural number, while the finite form of ‘be’ (*sono*) is 3rd person plural. Therefore there are multiple cases of checking here and the Number feature must survive at least some of these cases undeleted. Nonetheless, as stated above we remain neutral on the issue of whether subject agreement features form their own head or are associated with finite Tense.

2. Voice and Subject Agreement

As we outlined in the introduction, we consider Voice to have three principal properties: (i) the features REFL, A-CAUS, MIDDLE, PASS, ACTIVE seen in (2) and (4); (ii) these features can have exponents as e.g. clitic, affix, auxiliary, zero; and (iii) the licensing of the IA as e.g. *pro*, a weak pronoun, or a special/body-part pronoun. Properties (i) through (iii) may at first sight appear to be a rather *ad hoc* collection. However, they correspond very closely to standardly assumed properties of subject agreement.

Subject agreement is the locus of different features, such as Person, Number and Gender which, although conventionally labelled as ϕ -features for convenience, in fact are not a semantically natural class in that Person relates to reference/deixis, Number to quantification, while grammatical Gender is typically semantically vacuous. A further feature associated with subject agreement is Nominative Case, which like grammatical Gender is also semantically vacuous and therefore does not form a natural class with Person and Number. If we add EPP-features to this list, the set of AgrS features becomes even less natural as a class. In comparison, the features REFL, A-CAUS, MIDDLE, PASS and ACTIVE that we ascribe to Voice, along with the associated abstract Case features (including Accusative) are certainly no less of a natural class. What the two feature-sets seem to have in common is that they license arguments in different ways: in this sense the Case features are perhaps the most important ones.

Turning now to (ii), AgrS can be realised as a subject clitic (as in Northern Italian dialects, see for example Rizzi 1986a, Poletto 2000), as an affix (as in many Indo-European languages, among others), as an auxiliary (notably English *do*, which in the relevant syntactic contexts appears to function purely to carry subject agreement, see Chomsky 1957), or as zero (as in English, Chinese and Japanese among many other languages). Therefore, the different kinds of exponents we are attributing to Voice are nothing unusual; in fact, this appears to be the typical

cross-linguistic pattern of exponence of functional heads. One further kind of exponent which we will see below in the case of Chinese Voice (see section 9.2) is particles, i.e. invariant, phonologically reduced free morphemes. From a UG point of view exponence is random, but it is sensitive to hierarchical relations, notably asymmetric c-command and Relativised Minimality (for an approach along these lines, see e.g. Kallulli and Trommer 2011 on the realisation of non-active morphology in Albanian, in footnote 5). Moreover, languages tend to fall into well-known, rather loosely defined, morphological types, with most older Indo-European languages showing fusional inflection, Romance languages making extensive use of clitics and English showing a tendency to zero exponence (see footnote 12).²

Finally, let us consider point (iii): the licensing of the IA as e.g. *pro*, a weak pronoun, or a special/body-part pronoun. Clearly, AgrS plays a central role in licensing *pro*; this is the core of all accounts of the null-subject parameter stemming from Rizzi (1986b). Also, AgrS has been argued to license weak pronouns in various ways (see for example Cardinaletti and Starke 1999). Furthermore, to the extent that nominative pronouns are “special”, we can see that AgrS also shares this property with Voice, although the nature of the “specialness” is different, a matter that we believe largely reduces to different kinds of Case. All of these properties are subcases of a more general licensing ability which characterises these two heads. Of course, the fact that both heads can license structural Case (finite T licenses nominative, whereas active Voice licenses accusative) shows that both heads license DP arguments generally. The basic difference, then, is that AgrS licenses a subject (typically an external argument) DP, whereas Voice licenses an internal argument or a PP in its c-command domain. The reason for the difference is simply what we stated above: AgrS is high in the clause, possibly the highest head in the inflectional domain, while Voice is arguably the lowest head in the inflectional domain

² We will not speculate here on why functional heads are typically phonologically “small”, but see Roberts and Roussou (2003, chapter 5) for a proposal.

(or alternatively immediately below the inflectional domain as the highest head in the thematic domain; in section 8, we will consider the possibility that there is a separate “Voice field”). Our theory amounts to saying that Voice is a licensing head just like AgrS, its different behaviour follows from its different position in the clausal domain.³

Three things follow from these considerations. First, the simplest assumption is that Voice is universal; although our sample of languages is hardly typologically adequate, we will give evidence for Voice in Chinese and Austronesian, as well as Hebrew, in addition to several Indo-European languages from different branches (Albanian, English, German, Greek, Italian, Latin, Spanish).⁴ This approximates a “middle-way” approach of the kind advocated by Baker and McCloskey (2007). Second, the two licensing heads we have discussed seem to “bracket” the inflectional field in appearing very high (AgrS) and very low (Voice) in that field. It is not obvious why the clause structure should be organised in this way, but the observation seems clear and we suspect it is telling us something significant about Universal Grammar. Third, AgrS and Voice both exhibit Redundancy and Deficiency, as described by Onea et al. (2023): put simply, a feature is redundant if it is invisible at LF, and deficient, if it is invisible at PF (for details, see Kallulli and Roberts 2025b).

³ One further apparent difference between AgrS and Voice is that while the latter clearly licenses “downwards” (i.e., it asymmetrically c-commands its licensee) as we see in (1), (2) and (4), the former appears to typically license “upwards” (i.e., a DP in its specifier). However, preverbal subjects are frequently left-peripheral elements in many languages (Barbosa 1995), they often appear in “inverted” postverbal positions of various kinds in null-subject languages, they are always by definition postverbal in VSO and VOS languages, and frequently postverbal in V2 languages. We conclude that apparent upward licensing is not always a property of AgrS: further crosslinguistically variant factors seem to be at work in determining the nature of subject positions. These are beyond the focus of this article. Indeed, it seems fair to state that AgrS has crosslinguistic properties that are more variable than those of Voice.

⁴ See also Grestenberger (2021) on Vedic Sanskrit and Ancient Greek, to which we believe our analysis can be readily extended.

3. Non-active morphology and its (dis)guises

The study of both voice in the traditional sense and the proposals for the functional category Voice in recent syntactic theory have tended to focus on the active-passive alternation. Here, as we stated in the introduction, we will (following Kallulli 2007, 2013, 2021, Schäfer 2008, Harley 2013, Alexiadou et al. 2015, among others) look more widely in the attempt to advance a general theory of Voice, including Active Voice. In this section, we begin by illustrating the cross-linguistic similarities in marking of non-active Voice, something that does not appear at first sight to be found in English, although we argue that in fact it is, in the (dis)guise of systematic zero exponents. Finally, we will propose that Sportiche's (1996) framework for the analysis of clitic constructions provides the analytical template for our analysis of Voice.

To start with, it is well-known that what counts as a 'passive' form does not exclusively correspond to passive syntax the way this phenomenon is described for languages like English or German. That is, 'passive' form does not entail 'passive' syntax. For instance, while the Latin example in (5b) containing a verb which appears in what is traditionally known as the "passive conjugation" (glossed "non-active") as opposed to the "active conjugation" in (5a), is rendered in English as a passive, while in (6) it corresponds to an anticausative, and in (7) to a reflexive:

(5) a. *Auctor opus laudat.* (Ov. Pont. 3, 9, 9)

author_{NOM} work_{ACC} praise_{3S.PR.ACT}

'The author praises (his) work.'

b. *Laudatur Apronius a Trimarchide.* (Cic. Verr. 2, 3, 155)

praise_{3S.PR.ACT} Apronius_{NOM} from Trimarchides_{ABL}

'Apronius is praised by Trimachides.'

(6) *Africano illi superiori coronam sibi in convivio ad caput adcommodanti,*

Cum ea saepius rumperetur, (Cic. de orat. 2, 250)

while it_{NOM.F.SG} many.times break_{3S.SUBJ.IPFV.NACT}

P. Licinius Varus: “noli mirari” inquit “si non convenit; caput enim magnum est!”.

‘While Africanus, during the dinner, was putting back again on his own head the crown, since it (the crown) kept on breaking, P. Licinius Varus said: “You shouldn’t wonder that it doesn’t fit. In fact, you have a big head!”’

(7) *Abditur Orion.* (Cic, Arat. 462, 26)

hide_{3S.PR.NACT} Orion

‘Orion hides himself.’

The same phenomenon (i.e. syncretic non-active morphology) is also found in Albanian and Greek reflexives and anticausatives, as in (8) and (9):

(8) a. *Fëmija po lahet.* (Albanian)

child.the_{NOM} PROG wash.NACT.PR.3S

(i) ‘The child is washing itself.’ (reflexive)

(ii) ‘The child is being washed.’ (passive)

b. *To agori plithike (mono tu) / (apo ti mitera tu).* (Greek)

the boy washed.NACT.3S (alone his) / (by the mother his)

(i) ‘The boy washed himself.’ (reflexive)

(ii) ‘The boy was washed (by someone).’ (passive)

(9) a. *Vazoja* **(u)* *thye*.⁵ (Albanian)

vase.the_{NOM} NACT broke_{AOR.3S}

(i) ‘The vase broke.’ (anticausative)

(ii) ‘The vase was broken.’ (passive)

b. *To grama kaike* / **ekapse*. (Greek)

the sheet burned.NACT.3S / burned.ACT

(i) ‘The sheet burned.’ (anticausative)

(ii) ‘The sheet was burned.’ (passive)

Moreover, as Kallulli (2007, 2013) notes, in Albanian, Latin and Greek the *by*-phrase diagnostic cannot be applied to distinguish between passives and anticausatives because the relevant Preposition also means *from* (i.e. it may also introduce the external cause of an event), as illustrated in (5b) above and (10):⁶

⁵ Albanian employs three distinct means with a fixed distribution to build the non-active paradigm, namely: affix, as in (8a), clitic, as in (9a), and auxiliary choice (‘to be’). The distribution of non-active realization follows the pattern in (i):

- (i) If the clause contains Perfect: (Kallulli and Trommer 2011: 284)
 express non-active by choice of the auxiliary
 Else: If the clause contains Tense but not Aspect or Admirative:
 express non-active by an inflectional affix
 Else: express non-active by a clitic.

See Kallulli and Trommer (2011) and Kallulli and Roberts (in press) for analyses of these alternations in terms of Agree (and not, e.g., affix-hopping). More specifically, relying on a version of Agree which distinguishes valuation and interpretability, Kallulli and Trommer (2011) reduce the complexity of the Albanian non-active voice system to a very simple set of Agree and impoverishment relations, with the implicational choices in (i) reduced to local c-command (essentially Relativised Minimality): Perfect is realised when activated in preference to Tense or Clitic, since it is the closest c-commander of Voice; (non-Admirative) Tense is realised in preference to Clitic when Perfect is not activated, since it is the next-closest, and Clitic is realised when neither Tense nor Perfect is activated (Kallulli and Trommer 2011: 284-286). Kallulli and Roberts (in press) on the other hand, reduce Kallulli and Trommer’s (2011) version of Agree to just valuation of missing features.

⁶ An anonymous reviewer informs us that Hebrew shows the same pattern of syncretism between passive and anticausative, likewise disambiguated by the choice of preposition:

- (i) *he-esek* *nisgar al-jedej* *ha-rašujot*.
 the-business closed by the-authorities
 ‘The business was closed by the authorities.’
 (ii) *ha-xalon* *nisgar me-ha-ruax*.
 the-window closed from-the-wind
 ‘The window closed from the wind.’

The exponence of non-active Voice in Hebrew is through templatic morphology. We will say more about this in section 9.1 below.

(10) a. *Anna u dogj nga dielli mbi urë.* (Albanian)

Anna NACT burnt_{AOR.3S} by/from sun.the on bridge

(i) ‘Anna burned from the sun on the bridge.’

(ii) ‘Anna was burned by the sun on the bridge.’

b. *To grama kaike apo ti fotia.* (Greek)

the sheet burned.NACT by/from the fire

(i) ‘The sheet burned from the fire.’

(ii) ‘The sheet was burned by the fire.’

Similar syncretisms also exist in languages with no fully-fledged voice paradigms, such as Italian and German, which use a reflexive clitic or (a weak) pronoun instead of inflection:

(11) a. *Martina *(si) guarda allo specchio.* (Italian)

Martina REFL.3 watches in-the mirror

‘Martina watches herself in the mirror.’ (reflexive)

b. *Lamela *(si) mangerà domani.*

the apple REFL eat.3S.FUT tomorrow

‘The apple will be eaten tomorrow.’ (passive)

c. *Lo specchio *(si) è rotto.*

the mirror REFL.3 is.3S broken

‘The mirror has broken.’ (anticausative)

(12) a. *Ralf rasiert *(sich).* (German)

Ralf shaves REFL.3

‘Ralf is shaving/shaves (himself).’ (reflexive)

b. *Dieser Roman liest *(sich) gut.*

this novel reads REFL.3 well

‘This novel reads well.’ (middle)

c. *Die Tür öffnete *(sich).*

the door opened REFL.3

‘The door opened.’ (anticausative)

Moreover, both languages contain a class of verbs where, unlike in (13) and (14), the special (i.e. reflexive) morphology doesn’t seem to bear on argument-structure alternations. Hence, the ungrammaticality of (15b) and (16b) as opposed to the grammaticality of (13b) and (14b) must be due to a theta-criterion violation:

(13) a. *Martina si lava.* (Italian)

Martina REFL.3 washes

‘Martina washes herself.’

b. *Martina lava la camicia.*

Martina washes the shirt

‘Martina washes the shirt.’

(14) a. *Martina wäscht sich.* (German)

Martina washes REFL.3

‘Martina washes herself.’

b. *Martina wäscht das Hemd.*

Martina washes the shirt

‘Martina washes the shirt.’

(15) a. *Martina si arrabbia spesso.* (Italian)

Martina REFL.3 angers often

‘Martina often gets angry.’

b. **Martina arrabbia spesso Piero.*

Martina angers often Piero

(16) a. *Ich schäme mich.* (German)

I shame myself

‘I am ashamed of myself.’

b. **Ich schäme dich / (die) Eva.*

I shame you / (the) Eva

In fact, as Reuland (2009) remarks even for English: “[s]omething special must be said about [the paradigm in (18a,b) as opposed to (17a,b)]: Is it about binding or is it about argument structure?”

(17) a. John washed (himself).

b. John washed the child / us.

(18) a. John behaved (himself) / *us.

b. *John behaved the child.

The question then is what the role of the reflexive element in the (a) examples in (15a), (16a) and (18a) is. Kallulli (2013, 2021) argues that the reflexive element in these examples is the counterpart of non-active or passive morphology in the class of verbs known from traditional grammars of Latin as ‘deponent’ verbs, which are definable as verbs that lack an active conjugation and therefore only appear in the non-active (or so-called “passive” conjugation) but which do not have a passive meaning, as illustrated for Latin in (19b) as opposed to (19a), and for Albanian in (20a-e) as opposed to (20a’-e’).

(19)	<u>Pres.act.</u>	<u>Pres.pass</u>	(Latin)
a. alternating	<i>am-ō</i>	<i>am-or</i>	
	‘I love’	‘I am loved’	
b. deponent	—	<i>hort-or</i>	
		‘I encourage	
(20)	<u>Non-active</u>	<u>Active</u>	(Albanian)
a. <i>dergjem</i>		<i>a’ *dergj</i>	
	‘I linger’		
b. <i>përgjigjem</i>		<i>b’ *përgjigj</i>	
	‘I answer’		
c. <i>krenohem</i>		<i>c’ *krenoj</i>	
	‘I take pride in’		
d. <i>ligem</i>		<i>d’ *lig</i>	
	‘I weaken’		

e. *pendohem* e'. **pendoj*
 'I regret'

It is important to note that the morphological expression of non-active varies (dedicated non-active morphology, reflexive clitic/pronoun, auxiliary choice, etc), but we treat it as a syntactic category. In Albanian for instance the exponence of the Voice head REFL, A-CAUS, PASSIVE and MIDDLE can be any of the three means used to build the non-active (NACT) paradigm, namely: affix, clitic, auxiliary selection ('to be'), as we illustrate throughout (see also note 5). Therefore, cross-linguistically we may among other things expect null exponents, as is the case in English, for instance. Specifically, English has: (i) no (verbal) clitic or affix in reflexives, as in *John likes himself*; (ii) null deponent marking on the verb, as in *John absented himself*; (iii) no morphological marking of anticausatives, as in *The vase broke*; and (iv) no morphological marking of middles, as in *Bureaucrats bribe easily*. So, with the apparent exception of the participial passive (see section 7), English has null exponence of NACT (see Keyser and Roeper 1984, 1992 on abstract clitics in English).

Before proceeding with our analysis of each type of non-active Voice in turn, we first introduce the central ideas on clitics from Sportiche (1996). The reason for this is that Sportiche's general framework for the analysis of clitics provides, as we intend to show, a useful structural template for the analysis of Voice summarised in the introduction; see (1), (2) and (4).

Starting from the idea that both base-generation (Jaeggli 1980, Borer 1981) and movement analyses (Kayne 1975) of clitic constructions must be correct, Sportiche (1996) provides a structural analysis that combines both approaches. More specifically, he claims that all clitics are always base-generated as X^0 -elements of their own projections in whose specifier position they license a particular property/feature F; Sportiche in fact calls these projections "Voice

Phrases” but here we refer to them as Clitic Phrases (ClPs). These ClPs are located high in the clause, in a position higher than T/AgrS. For Sportiche, movement of an argument XP to SpecClP is motivated by the assumption of a general requirement for feature-licensing in Spec-Head configurations (we do not adopt this assumption here since our approach is Agree-based).

Sportiche makes a distinction between two types of clitics. The first type (typically accusative clitics) is operator-like; Sportiche argues that the operator-like property these clitics license is specificity in DPs. The second type (typically Romance dative clitics) are pure agreement markers, i.e. elements devoid of interpretive import (i.e. AgrIO heads in the sense of Chomsky 1995). As we have already mentioned, our analysis of Voice treats it as comparable to an Agr head and so analogous to Sportiche’s analysis of dative clitics. Below we will argue that Voice licenses oblique Case of various kinds; in substantive terms then our analysis dovetails very neatly with Sportiche’s in spite of the differences in technical implementation. Further, Sportiche proposes the following parameters:

(21) Clitic Parameters:

- a. Movement of the argument XP to SpecClP occurs overtly or covertly.
- b. The clitic head (Cl) is overt or covert.
- c. The argument XP is overt or covert.

These clitic parameters predict a full typology of clitic constructions, including what is most important for our purposes here, namely: (i) clitic doubling (as in e.g. Spanish, Romanian, Greek, Albanian), where an overt argument XP moves covertly to the Specifier of an overt clitic head; and (ii) scrambling in Dutch and German, where an overt argument XP moves overtly to the Specifier of a covert clitic head. In what follows, we will see that Voice heads typically instantiate the parametric options in (i) and (ii).

With this background in mind, we can now proceed to the analysis of the individual Voice patterns, beginning with reflexives.

4. Reflexives

An influential analysis which aims to capture the voice(-related) syncretism between reflexives, passives and anticausatives is the so-called “unaccusative” analysis of reflexives. Under this analysis, the source of the syncretism is a particular syntactic property, namely the absence of an external argument (Marantz 1981, 1984). Thus, modelling this insight in terms of Distributed Morphology, Embick (1997, 2004) argues that the syntax generates passives, reflexives and unaccusatives, each fully specified for distinct features, with the “u-syncretism” as he calls it, resulting from the realization of *v* in a particular structural environment, as in (22), where *-X* refers to the feature associated with non-active morphology:

$$(22) v \leftrightarrow v\text{-}X / \text{---} \text{No external argument}$$

By underspecifying *-X*, an account is thus provided for the common morphological form of all these constructions that otherwise have different syntactic and semantic properties. In other words, unaccusatives, passives and reflexives are not identical, but they share a common structure, namely the lack of an external argument; this is what the morphology is sensitive to, or reflects.

Given this background, the main question we want to focus on here is the following: Is (REFL/NACT) Voice always implicated in reflexivity? Our contention is that this is indeed the case. That is, (REFL/NACT) Voice is always implicated in reflexivity though its exponence can be null. Reasons to think that Voice is implicated in reflexive constructions bear on: (i) the

frequent cross-linguistic deployment of the same morphological markers for reflexives and (other) voice alternations, as we saw above, which in turn has motivated (ii); (ii) the “unaccusative analysis” of reflexives (Marantz 1981, 1984, Kayne 1993, Pesetsky 1995, Sportiche 1998, 2014, Embick 2004 *i.a.*); (iii) very similar locality conditions (which can be explicitly captured by subjecting both reflexives and NP traces to Principle A of Binding Theory), see (23) and (24); (iv) the cross-linguistically common subject-orientation of reflexives, while voice-alternations always implicate subjects/external arguments.

- (23) a. Mary was declared (Mary) bankrupt.
 b. *John was believed that (John) was bankrupt.
 c. John was believed (John) to be bankrupt.
 d. *John was believed Mary to like (John).

- (24) a. Mary has declared herself bankrupt.
 b. *John believed that himself was bankrupt.
 c. John believed himself to be bankrupt.
 d. *John believed Mary to like himself.

Thus, we maintain that the Latin non-active *-r* paradigms as in (19) are naturally associated with Voice (however exactly the Latin verb is built, on which see Calabrese (2021)); this naturally extends to (the exponents of) the Albanian and Greek non-active voice, as well as to so-called “SE” (as opposed to SELF-anaphors, cf. Reinhart and Reuland 1993 and below). In other words, simplex reflexive elements such as the Italian reflexive *si* merge in Voice and raise with the verb, just like non-active markers:

(25) a. *Abditur Orion.* (Latin)

hide_{3S.PR.NACT} Orion

‘Orion hides himself.’

b. *Gianni si ama / lava.* (Italian)

Gianni REFL love.3S / wash.3S

‘Gianni loves/washes himself.’

SELF-anaphors on the other hand are merged as true arguments, i.e. XPs; note the complementarity of SI/NACT with SELF-anaphors however:

(26) a. *Beni po lan veten / Anën.* (Albanian)

Ben PROG wash_{ACT.PR.3S} self_{ACC} / Anna_{ACC}

‘Ben is washing himself / Anna.’

b. **Beni po lahet veten / Anën.* (Albanian)

Ben PROG wash_{NACT.PR.3S} self_{ACC} / Anna_{ACC}

‘Ben is washing himself / Anna.’

(27) a. *Gianni lava se stesso / Anna.* (Italian)

Gianni washes REFL.3 self_{MASC.S} / Anna

‘Gianni washes himself / Anna.’

b. *Gianni si lava (*se stesso / Anna).* (Italian)

Gianni REFL.3 washes REFL.3 self_{MASC.S} / Anna

‘Gianni washes himself.’

(28) John is washing himself / Anna. (English)

We analyse these facts on the basis of the general cross-linguistic schema in (29), complemented by the dimensions of variation discussed below.

(29) General cross-linguistic schema for reflexives:

[VoiceP [Voice REFL] ... [VP V IA-pronoun)]]

REFL in (29) is the Voice-feature which marks the predicate as reflexive, in the sense that reflexivity needs to be licensed; our approach is thus similar in spirit to Reinhart and Reuland (1993) but differs notably in that we take reflexivity to be part of the Voice system.⁷ Its cross-linguistic exponence may involve an affix, a clitic, auxiliary choice (namely, ‘to be’), a zero morpheme, or, depending on the language, potentially some other linguistic means. For instance, in Italian REFL is the clitic *si* with the IA realised as *pro*. On the other hand, in Albanian the exponence of REFL can be any of the three morphosyntactic means used to build the non-active voice paradigm, namely an affix, as we already saw in (8a), a clitic, as in (30a),

⁷ Ahn (2015: 95f., 2016) also postulates a REFL Voice. However, his version of this idea is quite different from ours. In his system, REFL Voice has an EPP feature which attracts anaphors to its Specifier, although the lower copy is spelt out. Since, like us, Ahn assumes that Voice is outside the thematic domain, the only element which can function as an antecedent for the reflexive is the subject. Furthermore, this cannot be a derived subject, as such subjects depend on a different specification of Voice (e.g. passive, raising or non-active). This analysis derives the basic properties of anaphors in contexts of local subject-oriented reflexivity (i.e. canonical subject-oriented reflexives in active clauses). The representation of *Werner loved himself* is as in (i):

(i) [TP *Werner -ed* ... [PredP (*Werner*) Pred^o ... [VoiceP *himself* REFL ... [vP *Werner loved*
[vP *himself*(love)]]]]]

Here the external argument *Werner* raises from Spec,vP to SpecTP in the standard way (through SpecPredP as PredP may be a phase); the verb raises from V to v. Most importantly for our purposes *himself* raises to SpecVoiceP attracted by the EPP feature of Reflexive Voice. This movement guarantees the extrametricality of the reflexive, i.e. the fact it does not receive Nuclear Stress; instead, the Verb does (hence the accent on *loved*). In general, Nuclear Stress is assigned to the most deeply embedded category, and so normally the object of the transitive Verb is stressed (cf. *Werner loves Máry*), but movement of the reflexive has the consequence that the Nuclear Stress Rule does not apply to it (see Ahn 2015: 87-95 for details). Ahn’s account, unlike ours, is not grounded in a wider theory of Voice alternations, and assumes a rather different derived structure for reflexives from ours.

or BE, as in (30b):⁸

(30) a. *Beni *(u) la.*

Ben REFL wash

(i) ‘Ben washed himself.’

(ii) ‘Ben was washed (by someone).’

b. *Beni ishte larë.*

Ben was washed

(i) ‘Ben had washed himself.’

(ii) ‘Ben had been washed (by someone).’

English on the other hand has zero exponence for the Voice feature REFL (cf. Keyser and Roeper 1984, 1992), as in (31a), though given (31b), it stands to reason to state that REFL in this language can also be realised by a non-null exponent, namely the element *self-*, which at first sight seems to act as an incorporated noun.⁹

(31) a. John Ø loves/washes himself.

b. I heard that John self-harms.

Despite this variance of exponence within and across languages, of course REFL licenses the internal argument pronoun (IA-pronoun in (29)) in all cases, as depicted in (32) for Italian,

⁸ As discussed, all these examples also have a passive interpretation given the syncretic morphology, so in the context of reflexives only the (i)-readings are relevant.

⁹ We place Ø in (31a) in preverbal position since in English the lexical verb does not raise out of the vP and hence Voice precedes it.

A reviewer asks whether the analysis in (31a) is intended to carry over to indirect object reflexives in English as in: John sent himself a letter / John sent a letter to himself. We see no reason why this should not be the case.

Latin and English:

- (32) a. *Gianni si ama pro.*

Gianni REFL love.3S

‘Gianni loves himself.’

- b. *Abditur Orion pro.*

hide_{3S.PR.NACT} Orion

‘Orion hides himself.’

- c. John Ø hates himself.

- d. John self-harms *pro*.

The SELF-anaphor variant then clearly fits the schema in (29). As we will see below, this approach is fully in line with Sportiche’s parameters for clitics, which we are extending to Voice here as we outlined in the previous section.

Another dimension of variation in reflexives involves the realisation of licenser and licensee. Thus, in the examples above we see the realisation of either the licenser or the licensee, but in Greek and Albanian, both can -- indeed must -- be realised for the reflexive reading to obtain, as Embick (2004) discusses for Greek and as illustrated in (33) for Albanian and (34) for Greek.¹⁰ In fact this is to some degree possible even in English, as shown in (35).¹¹

¹⁰ The Albanian and Greek examples in (33) and (34) would be ungrammatical in the absence of non-active morphology, and have a passive interpretation in the absence of the ‘self’-reflexive element.

¹¹ An anonymous reviewer points out the following contrast as a problem:

- (i) John self-harms *pro* more than George self-harms *pro*. (Subject comparison)
- (ii) John self-harms *pro* more than John self-harms George. (Object comparison)

The reviewer points out that the object-comparison interpretation in (ii) is not available under ellipsis. If we assume the *self*-incorporation analysis, then *self* starts out in the object position and so *George* cannot occupy that position, and so the interpretation in (ii) is impossible. Notice that the operation of *self*-incorporation in English is analogous to the incorporation of *vetë* in (33) and *afto* in (34). Since as we have argued *self* is a noun, it falls under the generalisation in Kallulli (1997a,b) that bare (singular) nouns are semantically property-denoting expressions and syntactically lack a D-projection altogether, which enables them to undergo incorporation (see

Relatedly, we can have neither the licenser nor the licensee realised, as in (36).¹²

(33) *Jani u vetëshkatërrua.* (Albanian)

Jani NACT **self**destroyed.3S

‘Yani destroyed himself.’

(34) *O Yanis afto-katastrafike.* (Greek)

the Yani self-destroy.NACT.3S

‘Yani destroyed himself.’

(35) John *self-hated / ??self-promoted / ?self-nourished / self-selected / self-harmed

himself.

(36) John washes (himself).

also Van Geenhoven 1998). Incorporation of N to V may seem to be a problem for Roberts’s (2010) idea that only defective goals (i.e., heads whose features are a subset of those of the probe, i.e., the target of incorporation), may incorporate, but in semantic terms bare nouns share with verbs the property of being predicates (see again Kallulli 1997a,b).

¹² As (35) and (36) show, there is a degree of apparent lexical idiosyncrasy in English regarding which verbs allow *self*- and/or a reflexive object. It seems clear that Class I psych verbs (in the sense of Belletti and Rizzi 1988) do not allow *self*-, as *self-hate* in (35) and *self-like* in **Boris likes a lot* (in the interpretation ‘Boris likes himself a lot’), pointed out to us by an anonymous reviewer, show. Furthermore, as also pointed out to us by the same reviewer, while *self-control* and *self-defence* exist as nouns, the corresponding verbs *self-control* and *self-defend* are marginal (cf. ?*She self-controls all the time* and ?*She self-defends really well*). We suspect that there is ongoing change in this area in contemporary English relating to the ever-greater prevalence of zero-derivation. This is illustrated by the fact that, while the second author, a native speaker of a relatively conservative variety of British English has both versions of *I will address (myself to) this question*, his 24-year old daughter rejects the version with the reflexive. We, thus, obviously have change-in-progress here. Notice in this connection the existence of the nominal form *self-hate* (alongside *self-hatred*) which further corroborates our idea that zero-derivation is becoming more prevalent in contemporary English. Finally, the fact that the reflexive is optional with *behave* (as in *behave yourself*) but impossible with *misbehave* underlines the same tendency towards reduction of affixation. The current English tendency to zero exponence gives an impression of surface analyticity. However, when compared with the true, deep analyticity of Chinese (see Huang 2015), we see that English simply makes use of a great deal of zero suffixation while Chinese hardly has any suffixation. See Reuland (2011) for discussion of apparent lexical variation in *zich*-reflexives in Dutch.

We thus have an analogy with clitic doubling, as analysed in Sportiche (1996):

(37) [_{ClP} O(bject) Cl] [_{VP} V IA]]

Further developing and expanding on this analogy, we have the dimensions of variation in (21), which very closely parallel Sportiche’s (1996) Clitic Parameters, and which are repeated here for ease of reference as (38):

(38) Clitic Parameters:

- a. Movement of the argument XP to SpecClP occurs overtly or covertly.
- b. The clitic head (Cl) is overt or covert.
- c. The argument XP is overt or covert.

We uniformly account for the fact that though Greek and Albanian (unlike Spanish) don’t allow doubling of the *X-self* with overt REFL, there is nonetheless clitic doubling, as shown in (39), which we crucially interpret as the exponence of REFL.¹³ This is then a second novelty of our approach.

(39) a. *Fernando_i se_i / *lo lava [a si mismo]_i.* (Spanish)
 Fernando REFL/CL washes a himself

¹³ On the indexation in (39a,c) see the discussion in Anagnostopoulou and Everaert (1999); (39b) exactly parallels (39c) in this respect. Differently from these authors, however, we attribute the contrast between Albanian and Greek on the one hand versus Spanish on the other not just to properties of anaphora, but to properties of Voice and of the clitic systems across the two types of languages; for instance unlike Albanian and Greek, which are so-called “strong” PCC languages, Spanish is a “weak” PCC language (see Bonet 1991, Anagnostopoulou 2005), and these two things are connected, since we are tracing here a close connection between clitic and Voice systems inspired by Sportiche (1996).

b. *Ben_j e_{i/*j} do [veten_i (e vet_j)]_i.* (Albanian)

Ben CL.ACC loves self.the_{ACC} own

‘Ben loves himself.’

c. [*O Petros*]_j *ton_{i/*j} agapa_i [ton eaf_{to}_i tu_j]_i.* (Greek)

the Petros CL.ACC loves the_{ACC} self_{ACC} his

‘Petros loves himself.’

In turn, this is compatible with Woolford’s (1999) argument that object agreement is incompatible with anaphors, unless the agreement is a special anaphoric form. The structure for English, which has no overt clitics (but see Keyser and Roeper 1984, 1992), could then be as in (40), where *himself* is decomposed into the possessive pronoun *his* and the reflexive *self*; note that in English in all non-third-person X-self forms (i.e., *myself*, *yourself*, *ourselves*, *yourselves*) X is clearly a possessive pronoun, which is also the case for the third person singular feminine *herself*. The third person forms *himself*, *themselves* and *themselves* in Standard English deviate from this pattern, but many non-standard varieties have the forms *hissself* and *theirselves* (see also Helke 1979 and Ahn and Kalin 2018).

(40) John_j likes Ø_i [his_j self_i]_i.

Hence, another dimension of variation in reflexive constructions bears on the nature of the IA-pronoun, which as we have seen can among other things be *pro*, a weak pronoun (e.g. German *sich*, see Cardinaletti and Starke 1999),¹⁴ or a special/body-part pronoun (e.g., *X-self*); if the

¹⁴ In fact, German *sich* does not behave exclusively as a weak pronoun. As Ruigendijk and Schumacher (2020) point out, when *sich* is a logical argument it can carry stress, but with inherent reflexive verbs (e.g. *sich schämen* ‘to be ashamed of oneself’) it cannot. We interpret this to mean that *sich* does not obligatorily attach to a verbal head, unlike Italian *si* for example. We thank an anonymous reviewer for drawing our attention to this.

self-incorporation analysis suggested above for English is correct, it can also be an incorporating pronoun (i.e., a defective goal, see footnote 11). Taking stock, we can then revise (29) as in (41):

(41) [VoiceP [Voice REFL] ... [VP V IA-SELF/BODY(PART)]]

A further upshot of the analysis we have provided is that it accounts for selectional restrictions with pseudo-reflexives:¹⁵

(42) a. The ascetic_i injured himself_i / his_i body to hardship.

b. Mary_i exerted herself_i.

c. Mary_i exerted every ounce of her_i { energy / strength }.

(43) a. *The ascetic injured my body to hardship.

b. *Mary exerted every ounce of John's { energy / strength }.

Mandarin Chinese is another language which fits into the type with both licenser or licensee realised. As Li Zhen (personal communication) points out to us, research on reflexive verbs in Chinese is relatively limited, with scholarly work having primarily focused on the reflexive *ziji* 'self', paying less attention to *zi*-verb (i.e., the element *zi* combined with a verb, commonly referred to as a reflexive verb), but see Yu (1996), and Wong (2021). However, Li Zhen further points out to us that some reflexive verbs can take an overt object coreferential with the

¹⁵ As mentioned, Kallulli (2013, 2021) takes pseudo-reflexives to be the counterparts of deponent verbs. See these works also for arguments that deponents are truly non-agentive and therefore structurally unaccusative. Following those arguments, we take examples like (42) to be able to show secondary agentivity effects on a par with examples such as *John deliberately fell asleep*.

subject/clitic *zi*. In other words, both the licenser and the licensee appear. For example, for him reflexive verbs such as *zi-xing* ‘self-reflect’ and *zi-hui* ‘self-destroy’ can take an object in Mandarin Chinese, as shown in (44) and (45), “although the versions without the reflexive *ziji* could be more natural”.

(44) *Lisi zhende shi zai zi-hui (ziji).*

Lisi really be PROG self-destroy self
 ‘Lisi is really destroying herself/himself.’

(45) *Lisi jingchang zi-xing (ziji).*

Lisi often self-reflect self
 ‘Lisi often reflects herself/himself.’

To conclude, if we treat reflexives as involving Voice, they fit into a consistent, independently attested pattern of cross-linguistic variation in properties of functional heads. In the next section, we turn to the analysis of anticausatives.

5. Anticausatives

Consider the examples in (46), which illustrate anticausatives:

(46) a. *Lo specchio si è rotto.* (Italian)

the mirror REFL.3 is broken
 ‘The mirror has broken.’

- b. *ea saepius rumperetur.* (Latin)
 it_{NOM.F.SG} many.times break_{3S.SUBJ.IPFV.NACT}
 ‘it breaks many times’
- c. *The window broke.* (English)
- d. *Vazoja u thye.* (Albanian)
 vase.the_{NOM} NACT broke
 ‘The vase broke.’
- e. *Die Tür öffnete sich.* (German)
 the door opened REFL.3
 ‘The door opened.’

In anticausatives, we observe the same morphosyntactic variants as for the reflexives. The general cross-linguistic schema for anticausatives is given in (47):

(47) General cross-linguistic schema for anticausatives:

[VoiceP [Voice A-CAUS] ... [VP V (IA)]]

Here A-CAUS is the Voice-feature which marks the predicate as anticausative. The exponence of A-CAUS is as for reflexives, i.e. it may be a clitic/weak pronoun, an affix, an auxiliary or zero. In Albanian, again as in the case of reflexives, we find three patterns of exponence: a clitic (9a), affix (48a), or the auxiliary BE (48b):

(48) a. *Çelësi thyhej sa kthente bravën.*

key.the_{NOM} break.NACT.P.3S when turned lock

‘The key would break as soon as it turned the lock.’

b. *Vazoja ishte / *kishte thyer.*

vase.the_{NOM} was / had broken

‘The vase had (been) broken.’

These exponents are determined as described in note 5 (see also Kallulli and Roberts, in press).

Null exponence of A-CAUS is also possible, as in English (46c) above.¹⁶

In the case of anticausatives, unlike reflexives, the question of the representation of the external argument arises. The argument in the *from*-phrases in (49) clearly corresponds to the Causer (only the (i)-readings are relevant in (49c,d)):

(49) a. *Lo specchio si è rotto dalla pressione.* (Italian)

the mirror REFL.3 is broken from-the pressure

‘The mirror broke from the pressure.’

b. *The window broke from the heat.* (English)

¹⁶ An anonymous reviewer reminds us that anticausatives do not always show a SE-reflexive in French (Ruwet 1976), Italian (Burzio 1986), or Dutch (Everaert 1986, Reinhart et al. 2016). The same is true regarding mediopassive morphology in Greek and Albanian, indicating that these alternations are independent of the realisation of non-active morphology. See Alexiadou et al. (2015), Schäfer (2024), Martin et al. (2025) for discussion and analysis. The latter relate the apparent lexically-governed idiosyncrasies to a combination of a human event participant and cognitive preferences of a pragmatic nature, e.g. “an agent bias”, along with the ambiguity of SE as a reflexive marker and an anticausative marker. These tendencies arise then as a consequence of the strong cross-linguistic tendency for uniform morpho-phonological marking of non-active Voice. The fact that similar Voice “gaps” are also found in middles across languages (e.g., Albanian, Greek, Dutch) underlines this conclusion; see also footnote 17. These languages differ from English because English lacks an overt SE morpheme, therefore we find systematic zero exponence of anticausatives across the board in English; this is a typological difference between English and the other languages considered here (see also footnote 12).

c. *Anna u dogj nga dielli mbi urë.* (Albanian)

Anna NACT burnt.AOR.3S by/from sun.the on bridge

(i) ‘Anna burned from the sun on the bridge.’

(ii) ‘Anna was burned by the sun on the bridge.’

d. *To grama kaike apo ti fotia.* (Greek)

the sheet burned_{NACT} by/from the fire

(i) ‘The sheet burned from the fire.’

(ii) ‘The sheet was burned by the fire.’

e. *Die Tür öffnete sich durch den Luftzug.* (German)

the door opened REFL.3 from the draft

‘The door opened from the draft.’

Since it bears the Causer role, we could treat the *from*-phrase as the external argument (see Kallulli 2007). We return to this question in section 8, meanwhile we attribute the representation in (50) to anticausatives where the *from*-phrase is present:

(50) [VoiceP [Voice A-CAUS] ... *from*P ... [VP V (IA)]]

Here Voice’s A-CAUS feature licenses the *from*P, which arguably intervenes between Voice and the IA. Where A-CAUS has an overt realisation (i.e. in all the languages under consideration except English), realisation of the *from*-phrase implies double exponence of the A-CAUS feature, again in line with Sportiche’s approach to clitic-doubling outlined in section 3.

In the next section, we turn to middles.

6. Middles

Middles resemble anticausatives and indeed passives in that the notional internal argument, typically a theme/patient, appears in the surface subject position. A salient property of middles which contrasts clearly with anticausatives is the fact that they are non-eventive, as illustrated in (51):

- (51) a. *Questi libri si leggono facilmente.* (Italian)
 these books REFL.3 read.3PL easily
 ‘These books read easily.’
- b. *Bureaucrats bribe easily.* (English)
- c. *Dieser Roman liest sich gut.* (German)
 this novel reads REFL.3 well
 ‘This novel reads well.’
- d. *Ky libër lexohet kollaj.* (Albanian)
 this_{nom} book read_{NACT.PR.3S} easily
 ‘This book reads easily.’

Middles show the same morphosyntactic variants as reflexives and anticausatives discussed in the previous sections.¹⁷ This is summarised in (52):

(52) General cross-linguistic schema and dimensions of variation for middles:

[VoiceP [Voice MIDDLE] ... [VP V (IA)]]

¹⁷ As we discussed for anticausative in the previous section, this includes Voice “gaps”, i.e., zero-exponents even in languages which otherwise have clitic, inflectional or periphrastic realisation of middles; see e.g. Broekhuis et al. (2016) on the alternation between clitic and zero in Dutch (brought to our attention by an anonymous reviewer).

Here MIDDLE is the Voice-feature which marks the predicate as a middle. The exponence of MIDDLE is as for reflexives and anticausatives (affix, clitic, auxiliary or zero). Again, Albanian has an affix (51d), a clitic (53a) or the auxiliary BE (53b) (only the (i) readings are relevant here):

(53) a. *Ky libër u lexoka kollaj.*

this_{NOM} book NACT read_{ADM.PR.3S} easily

(i) ‘Hmm, so this book reads easily.’

(ii) ‘Hmm, so this book is easy to read.’

b. *Ky libër qenka / *paska lexuar kollaj.*

this_{NOM} book be_{ADM.3S} / *have_{ADM.3S} read easily

(i) ‘Hmm, so this book has read easily.’

(ii) ‘Hmm, so this book has been easy to read.’

Finally, null exponence is also possible, as in English (51b) above.

As with anticausatives, a question arises concerning the representation of the external argument in middles. A characteristic feature of middles is that they have a dispositional reading (Condoravdi 1989, Lekakou 2005), implying that the clause is stative, the subject often is a generic bare plural, and an evaluative adverb characterising the property (canonically *easily*) appears (see Cinque 1999: 101-3, on Voice and the middle adverb). This adverb has an Experiencer argument, which may be realised as a *for*-phrase or its equivalent, illustrated in (54):¹⁸

¹⁸ Italian appears to be an exception to the cross-linguistic pattern observed in (54):

(i) **Questi libri si leggono facilmente per noi / gli alunni.*
these books REFL,3 read.3PL easily for us / the pupils

- (54) a. *This book reads easily for me.* (English)
- b. *?Dieser Roman liest sich leicht für mich.* (German)
- this novel reads REFL.3 easily for me
- ‘This novel reads easily for me.’
- c. *Physik-Bücher lesen sich leicht für Mathematiker...* (German)
- physics books read.3PL REFL.3 easily for mathematicians
- ‘Physics books read easily for mathematicians... (but not for linguists).’
- d. *Ky libër u lexoka kollaj (edhe) për mua.* (Albanian)
- this_{NOM} book NACT read_{ADM.PR.3S} easily also for me
- ‘Hmm, this book reads easily (also/even) for me.’

Hoekstra and Roberts (1993) and Stroik (1999) propose that the *for*-phrase introduces an Experiencer external argument. That this argument is an Experiencer and not a Benefactive can be seen from two tests. First, Experiencers can only be sentient beings while Benefactives can be inanimate. This can be seen in the following contrast, where *taste* in (56b) is unambiguously a psych verb with an Experiencer internal argument:

- (55) a. This kind of water is good for the kids / the lawn.
- b. This kind of water tastes good for the kids / *the lawn.

Here the *per*-phrase only has the “according to” interpretation; the external argument is interpreted as arbitrary (Leonardo Russo-Cardona, p.c.). It is not clear why Italian does not fit the cross-linguistic pattern. The *tough*-movement example corresponding to (i) is grammatical with a *per*-phrase corresponding to the external argument:

- (ii) *Questi libri sono facili da leggere per noi / gli alunni.*
- these books are easy to read for us / the pupils

We have no explanation for these facts at present.

Second, with many verbs the Experiencer can be introduced by *to* as well as *for*, whereas for Benefactives this is not the case, as can be seen in (56):

(56) This kind of water tastes good to the kids / *the lawn.

If this is correct, then we should replace (52) with (52'):

(52') [VoiceP [Voice MIDDLE] ... [AdvP ADV *for* P] ... [VP V (IA)]]

In (52'), the MIDDLE feature of Voice licenses the *for*-phrase. As with the anti-causatives discussed in the previous section, we can treat the languages in which MIDDLE Voice has an overt morphosyntactic realisation as double exponence, a form of clitic-doubling once more. English once again has null exponence of MIDDLE Voice, seen in (51b).

We mentioned above that middles are characterised by a property (or dispositional) reading. We suggest that this arises because the external argument of a middle is an Experiencer. Given that, like much recent work on voice alternations (for overview and references see Alexiadou, et al. 2015), we are not assuming that external (or other) arguments of predicates are lexically determined, it is quite possible to suppose that otherwise canonical agentive predicates can be interpreted as having Experiencer external arguments (in terms of Reinhart's 2002 Theta System, Experiencers and Agents share the +m feature; see also Reinhart et al. 2016, and Kallulli 2006). This means that the middle predicates share important properties with the 'fear' class of psych verbs (Belletti and Rizzi's 1988 Class I psych verbs), which are statives with Experiencer external arguments. These verbs also express dispositions. Furthermore, subject Experiencers introduce a path either as a goal or source unlike non-experiencer subjects (see Speas 1990 and Landau 2010). Speas (1990) illustrates this with the following examples:

(57) a. I got angry but it went away.

b. ??I laughed but it went away.

We see the same with Class I psych verbs:

(58) a. Mary used to love John but it went away.

b. ??Mary used to kiss John but it went away.

Now compare the following contrast between Benefactive and Experiencer *for*-phrases:

(59) a. ?Ancient Greek used to read easily for me but it went away.

b. *This water used to be good for my lawn but it went away.

Thus, we see that the *for*-phrase in middles has a further Experiencer property. We conclude that the effect of MIDDLE Voice is to cause the external argument to become an Experiencer argument, realised overtly as the *for*-phrase associated with the adverb. In Albanian and elsewhere (e.g., Slavic) dispositional and affective dative subjects are found, which in Albanian employ non-active morphology and in Slavic are marked by reflexives; see Kallulli (2006):

(60) *Benit i hahej një mollë.*

Ben_{DAT} CL.DAT.3S eat_{NACT.P.3S} an apple

‘Ben felt like eating an apple.’

There is a clear parallel between examples like (60) and the English middle construction (cf.

These apples cook easily for Benny). We therefore suggest that middles arise when an agentive, eventive transitive verb takes an Experiencer external argument instead of an Agent. Moreover, to the extent that canonical eventive transitive verbs have affected internal arguments (Roberts 1987), the conversion of such verbs to middles removes the affectedness interpretation exactly as is the case with Class I psych verbs (e.g., *snakes* is not an affected argument in *John fears snakes*). Since no verb can have two external arguments, only one of the two options can appear with a given occurrence of a verb of this type.

7. Passives

In this section we bring passive constructions into our general proposals, although we will leave certain questions concerning the cross-linguistic morphological properties of passives open.

Passives are illustrated for Italian, Latin, English and German in (61). In (61a), we give both the participial passive and “mediopassive” *si* of Italian. It is of course the latter that shows syncretism with the other types of non-active constructions, including reflexives, that we have been concerned with here. In (61b), we see the morphological passive of Latin. In perfective tenses, Latin has a periphrastic participial passive (see (65b) below). (61c, d) illustrate the English and German participial passive, which have different auxiliaries (German also has a ‘be’-passive, but this is usually considered to be an adjectival passive, see Alexiadou et al. 2015):

- (61) a. *Questi libri sono stati venduti / si sono venduti.* (Italian)
 these books are been sold / REFL.3 are sold
 ‘These books have been sold.’

b. *Laudatur Apronius a Trimarchide.* (Latin)

praise_{3S.PR.NACT} Apronius_{NOM} from Trimarchides_{ABL}

‘Apronius is praised by Trimachides.’

c. *The bureaucrats were bribed.* (English)

d. *Dieser Roman wurde gelesen.* (German)

this novel was read

‘This novel was read.’

In (62) and (63), we see the Albanian and Greek passives. In all these examples, we have syncretic non-active voice morphology, which, alongside the passive interpretations (boldfaced here), also allows a reflexive interpretation in (62) and an anticausative interpretation in (63):

(62) a. *Fëmija po lahet.* (Albanian)

child.the_{NOM} PROG wash.NACT.P.3S

(i) ‘The child is washing itself.’ (reflexive)

(ii) ‘The child is being washed (by X).’ (passive)

b. *To agoriplithike (mono tu) / (apo ti mitera tu).* (Greek)

the boy washed.nact,3s (alone his) / (by the mother his)

(i) ‘The boy washed himself.’ (reflexive)

(ii) ‘The boy was washed (by X).’ (passive)

(63) a. *Vazoja *(u) thye.* (Albanian)

vase.the_{NOM} NACT broke.AOR.3S

(i) ‘The vase broke.’ (anticausative)

(ii) ‘The vase was broken.’ (passive)

- b. *To grama kaike.* (Greek)
- the sheet burned.NACT
- (i) ‘The sheet burned.’ (anticausative)
- (ii) ‘The sheet was burned.’ (passive)**

Here the cross-linguistic pattern is as shown in (64):

(64) Cross-linguistic pattern:

[VoiceP [Voice PASSIVE] ... *by*P ... [VP V (IA)]]

Here the *by*-phrase is licensed by PASSIVE Voice, analogously to the *from*-phrase in anticausatives and the *for*-phrase in middles. However, the cross-linguistic pattern of exponence is different from what we saw above with anticausatives and middles in that English, German, Italian, Albanian (subject to tense/aspect distinctions; see Kallulli and Trommer 2011, Kallulli and Roberts, in press) and Latin (in perfect tenses) all have participial passives:

- (65) a. *Beni ishte larë.* (Albanian)
- Ben_{NOM} was washed
- (i) ‘Ben had washed himself.’
- (ii) ‘Ben had been washed (by someone).’**
- b. *A me Lesbia amata mea est.* (Latin, Catull. 87, 2)
- from me Lesbia_{NOM} love_{PST.PTCP.NOM.SG.F} mine_{NOM.SG.F} is
- ‘My Lesbia has been loved by me.’

Where the exponence of passive involves an auxiliary and a participle, we have either the structure in (66a) or that in (66b):

- (66) a. [VoiceP [Voice BE] ...[PrtP V+Prt [VP (V) (IA)]]
 b. [AuxP BE [VoiceP [Voice V+Prt] ...[VP (V) (IA)]]

The analysis of Albanian in Kallulli and Trommer (2011) and Kallulli and Roberts (in press) favours (66b) if AuxP is interpreted as PerfectP, since the activation of this projection entails the presence of an auxiliary (see note 5, (i)). The structure (66b) is also consistent with Harwood's (2013) analysis of the English auxiliary system. Hence, we tentatively favour this structure for participial passives, but we do not exclude the possibility of cross-linguistic variation, such that (66a) would be the correct structure in some languages; in the next section we will see an attractive and interesting consequence of placing BE above VoiceP, as in (66b).

In non-perfective tenses in Latin, generally in Greek and, subject to tense/aspect conditioning, in Albanian, passive is realised by a dedicated morpheme, shown in (67) (only the (ii) readings are relevant in (67b,c)):¹⁹

- (67) a. *Laudat-ur Apronius a Trimarchide.* (Latin)
 praise_{3S.PR.NACT} Apronius_{NOM} from Trimarchides_{ABL}
 'Apronius is praised by Trimarchides.'
 b. *Fëmija po la-h-et.* (Albanian)
 child.the_{NOM} PROG wash.NACT.PR.3S
 (i) 'The child is washing itself.' (reflexive)
 (ii) 'The child is being washed.' (passive)

¹⁹ Calabrese (2021:8), note 16, says that *-u-* in Latin is epenthetic; this also holds for *-h-* in Albanian in (67b).

c. *To agoripli-thike* (mono tu) / (*apo ti mitera tu*). (Greek)

the boy washed.NACT.3S (alone his) / (by the mother his)

(i) ‘The boy washed himself.’ (reflexive)

(ii) ‘The boy was washed.’ (passive)

The Verb is attracted to Voice here too (and possibly moves further up in the clausal spine, although this requires further demonstration, especially for Albanian); the difference with periphrastic passives lies in the presence of an auxiliary in a higher functional head, which bears Tense and subject-agreement features.

Of course, one can ask why we see the alternation, within and across languages, between periphrastic passive and the other syncretic non-active heads. To put it another way, why, with the exception of Albanian-like languages (see again Kallulli and Trommer 2011, and Kallulli and Roberts, in press), do we not see periphrastic reflexives, middles or anti-causatives?²⁰ One possibility is suggested by both Albanian and Latin, which have a periphrasis in the perfect (Albanian) or the perfective forms (Latin): it may be that the participle originates diachronically as denoting a resultant state, which develops into a perfect(ive) and thence into a generalised passive, at least in English, German and Italian among the languages we have considered here. Reflexives, with their lack of sensitivity to aspect, anticausatives, which tend to have an inchoative reading, and middles, which are purely stative, are all incompatible with the participial resultant-state interpretation; this may go some way to accounting for why these constructions do not originate as a periphrasis and why the participial periphrasis does not generalize to them. Formally, this could be captured in terms of a requirement that in each of these constructions, the lexical Verb must access the aspectual field, either through movement

²⁰ We take the notion of periphrasis to crucially involve auxiliaries, therefore we do not consider *self*-expressions in reflexive constructions as examples of periphrasis.

or Agree, and this is not possible if an auxiliary is present since an auxiliary will intervene between the position of the participial lexical Verb and the aspectual field. But we leave this as a tentative speculation for now. In the next section we will propose a more “cartographic” approach to Voice which may also shed light on this issue.

8. Summary: What is Voice?

In the preceding sections, we have seen the following scenarios involving non-active Voice:

- (68) a. [VoiceP [Voice REFL] ... [VP V IA-self/body(part)]]
 b. [VoiceP [Voice A-CAUS] ... *from* P ... [VP V (IA)]]
 c. [VoiceP [Voice MIDDLE] ... [AdvP ADV *for* P] ... [VP V (IA)]]
 d. [VoiceP [Voice PASSIVE] ... *by* P ... [VP V (IA)]]

To complete the picture, we now add Active Voice:

- (69) [VoiceP [Voice ACTIVE] ... [VP [VP V IA]]

So, we observe five different kinds of Voice. The obvious question at this point is what underlies these alternations. For the anticausatives, middles and passives, i.e. (62b, c, d), we propose abstract oblique Cases: ablative (‘from’), dative / benefactive (‘for’), instrumental / ergative (‘by’).²¹ In Greek, Albanian and Latin, there is syncretic realisation of ablative and

²¹ An anonymous reviewer asks how the prepositions that we have analysed here as oblique Case markers are related with other uses of these elements (e.g., benefactive *for*, locative *from* and *by*). We treat this as a diachronic relation. Synchronically there is no relation; these are functional prepositions derived from more “lexical” prepositions by normal grammaticalisation processes, a kind of “semantic bleaching” (see Roberts and Roussou 2003, chapter 5 for a general account of semantic bleaching).

dative, meaning that the same preposition appears in passives and anticausatives. Of course, case syncretism is a natural and expected process.

In these terms, ACTIVE Voice fits neatly into the picture, in that it licenses an Accusative IA. In general, Case-licensing categories license a single Case (e.g. finite Tense licenses Nominative and no other Case, etc). So, if Voice can license exactly one Case, then we understand why the IA cannot be accusative in (68b,c,d) and must therefore move. Extending this to the reflexives, since reflexive Voice licenses *self*, then we conclude that reflexive marking is a kind of Case (as independently proposed by Pesetsky 2011).

Our account leads to a new conclusion regarding the nature of syncretic non-active morphology: it marks Voice as a licenser of oblique Case. This distinguishes our approach from all previous analyses (but see note 22), including that proposed by Alexiadou et al. (2015).²² In terms of Redundancy and Deficiency, Case is deficient since it often lacks PF exponence, and redundant since it is uninterpretable.

One question that we have systematically left open in the foregoing is the first-merged position of the oblique argument introduced by the various prepositions (i.e., the passive *by*-phrase, the anticausative *from*-phrase, etc.). There are three reasons why this is central to the analysis of non-active voice. First, the optional presence of such PPs indicates, at the very minimum, that a notional external argument can be structurally present as the complement of the preposition. This becomes particularly clear if we consider the evidence that the DP in the

²² Alexiadou et al. (2015) present an analysis of non-active constructions across several languages. Unlike in our approach, they treat the passive in English and German as involving a Passive head which selects Voice bearing a D-feature representing the external argument. Passive selects VoiceP and binds the external argument (cf. also Bruening 2013). For passives in Greek and similar languages, on the other hand, these authors propose a structure without a Passive head, and they label Voice “Voicemiddle”, which subsumes several morphologically syncretic Voice heads. This kind of Voice is also found in “marked anticausatives” in Greek (i.e., anticausatives bearing non-active morphology), as well as in reflexives. On the other hand, they propose a different structure for “SE-reflexives” and SE-marked anticausatives in Romance and Germanic. We consider our approach to be superior for two reasons. First, because it provides a uniform structure for reflexives and all kinds of non-active voice, and second, because the account extends naturally to English, as we have shown (cf. also Legate et al. 2020, who also do away with an extra Passive projection above Voice).

passive *by*-phrase can bear exactly the same compositional thematic role as the active external argument, since in each of these examples the argument in (70) the DP in the *by*-phrase matches exactly the surface subject argument of the corresponding active in terms of the semantic role borne (Roberts 1987: 27):

- (70) a. A baseball was thrown by Fernando.
b. Support was thrown behind the candidate by the CIA.
c. The match was thrown by the prize-fighter.
d. The party was thrown by the department.
e. A fit was thrown by the countess.

Second, when the PP is absent, the corresponding argument is still understood, and in fact is usually interpreted as an indefinite of one kind or another, as in the following examples.

- (71) a. A baseball was thrown.
b. The ice melted.
c. Bureaucrats bribe easily.

Hence, (71a) means ‘Someone threw a baseball’, (71b) means ‘Something caused the ice to melt’, and (71c) means ‘It is easy for one to bribe bureaucrats’. Such indefinite arguments are referred to as implicit arguments. We briefly return to the question of implicit arguments in the conclusion; on the nature of these implicit (external) arguments, see Collins (2024) and references therein on passives, Kallulli (2007) on anticausatives, and Hoekstra and Roberts (1993) on middles.

Third, it was originally proposed by Perlmutter and Postal (1977) that passives require an external argument. Hence, while some languages, e.g. German, allow passives of unergative

intransitives as in (72a), passives of unaccusatives, as in (72b), are not allowed.

(72) a. Es wurde getanzt.

it was danced

‘There was dancing.’

b. *Es wurde angekommen.

It was arrived

Perlmutter and Postal formulated this as a universal constraint on passives. This implies that, in addition to passives of unaccusatives, passives of passives and passives of raising verbs are disallowed. Various apparent counterexamples to this generalisation, notably from Turkish, Lithuanian and Sanskrit, have been observed, but see Legate et al. (2020) for arguments that the constructions in question are passives of impersonals, passives of evidentials or simple unaccusatives. Furthermore, neither unergative nor unaccusative intransitives show either the anticausative or the middle alternation:

(73) a. *It was danced from the catchy music.

(Intended: The catchy music caused people to dance.)

b. *It arrives on time easily for trains like that.

(Intended: It is easy for trains like that to arrive on time.)

Three different ways of analysing the behaviour of these implicit (external) argument in passives have been proposed. One proposal is to treat them as structurally represented but simply not promoted from their first-merged, vP- or VoiceP-internal position. This approach is advocated by Collins (2005, 2024), Roberts (2019), Angelopoulos et al. (2020). It has the

immediate advantage of accounting for the first property above -- the fact that the *by*-phrase faithfully tracks the external argument of the active -- because on this view it *is* the external argument. This approach also satisfies a strong version of Baker's (1988) UTAH.

Since, on this view, the first-merged position of the external argument asymmetrically c-commands that of the internal argument, the latter cannot move across the former to SpecTP without incurring a violation of relativised minimality. This led Collins (2005) to propose that the VP, or some non-argumental category containing the internal argument, raises across the external argument without incurring a minimality violation. The internal argument is then raised to SpecTP from the moved category. This kind of derivation is known as smuggling.

If we take this option, then in all non-active constructions the relevant PPs can be seen as actually instantiating the external arguments (see e.g. Kallulli 2007), but we have to explain then why the external argument is not marked as reflexive in (68a) and why it is not marked Accusative in the active (with the internal argument presumably raising to SpecTP for Case).²³ Concerning the latter point, Roberts (2019) suggests inheritance of Case features from Voice to *v* in the active, thus ensuring that the internal argument is Accusative. For reflexives, adopting the unaccusative analysis as in section 4 neatly unifies reflexives with the other non-active Voices and this problem does not arise.

A second way of handling the properties of the implicit (external) argument is to treat the relevant head (i.e., in our terms Voice), as instantiating this argument without it being projected in a Specifier position. Embick (2004), Legate (2012), Bruening (2013), Alexiadou et al. (2015) and Legate et al. (2020) all adopt variants of this approach, as, in an earlier framework, did Baker et al. (1989). This approach requires a special statement for the relation between Voice

²³ An apparent counterexample comes from Albanian, where anaphors may bear overt nominative case in non-active constructions (Williams 1988, Massey 1992, Anagnostopoulou and Everaert 1999). According to the native intuitions of the first author, these nominative anaphors are actually logophors. Space limitations prevent us from entering into a detailed analysis of these data here, but see Massey (1992) for relevant discussion.

and the *by*-phrase and the other PPs in the other non-active constructions, which are typically treated as adjuncts. However, the approach satisfies a weak version of UTAH (but one arguably compatible with bare phrase structure in not taking the distinction between heads and XPs as primitive).

A further possibility is to take the implicit (external) argument to be first-merged in SpecVoiceP. In that case, in all the non-active Voices this argument fails to be structurally realised (although it can nonetheless be interpreted as an implicit argument), with the PP adjuncts Case-licensed by Voice.

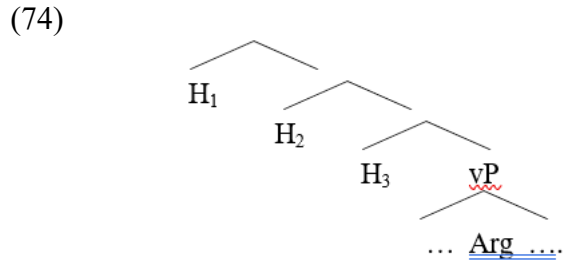
We do not choose among these options here, not least owing to the fact that the two authors have differing views on this matter and on the nature of implicit arguments; see Kallulli (2007) and Roberts (1987, 2019).

A further question prompted by an anonymous reviewer that arises in connection with (68) and (69) is whether the distinct types of Voice are actually structurally, rather than simply featurally, distinct. Could we arrange the different features of Voice in a hierarchy, or indeed could we propose, in the spirit of syntactic cartography, a “Voice field” in between the inflectional field and the argument-structure domain headed by the highest *v*, heading the highest vP/VP-shell, and introducing the highest argument?

Here we just sketch an answer to this question, since there is rather little direct empirical evidence for such a hierarchy. Nonetheless, the idea has a certain naturalness and is worth considering. Thus, instead of positing up to five different feature-specifications of Voice as in (68, 69), let us instead consider how the different kinds of Voice might be hierarchically arranged as distinct heads.

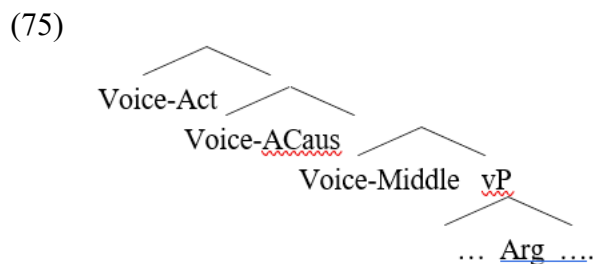
The first point which needs to be dealt with is the complementary distribution, or near-complementary distribution, of the features in (68) and (69). Middles, anticausatives and actives are clearly in complementary distribution. We can account for this in hierarchical terms

by arranging the relevant heads in a hierarchy such that each head is an asymmetric c-command relation with the other two and then apply Featural Relativised Minimality (as in Rizzi 2013) to the potential Agree relations such that the higher heads can only Agree with a vP-internal argument if the others are “inert”, i.e. their features, or the actual head itself, are absent. Schematically, then, we have the situation in (74):



Here, given the standard non-intervention condition on Agree (which can be seen as Featural Relativised Minimality), assuming that the features of H₁, H₂ and H₃ are all in the same class, i.e. Voice or Case features, then if H₃ has a feature probing Arg, this will prevent H₂ and H₁ from having such a feature, and similarly for H₂ in relation to H₁. The question is, though, which of the heads here is which?

We propose that H₃ is the Middle head, since middles directly alter the thematic structure of vP, pre-empting the possibility of an agentive vP; given our proposals in Section 6, the argument directly probed by the Middle head must be an Experiencer. Similarly, we propose that H₂ is anti-causative, since this head requires a canonically transitive vP complete with an agent argument, as we saw in Section 4. By elimination, then, H₁ is the Active head, licensing a vP-internal argument as Accusative. So we have the partial structure in (75):



Unlike Voice-ACaus and Voice-Middle, Voice-Act is compatible with other internal

arguments in vP (we discussed the question of external arguments above), hence it can only be active when Voice-ACaus and Voice-Middle are “inert” (either structurally absent or featurally defective); another way to put this might be to say that Voice-Active is indifferent to the presence of an Applicative head in or immediately dominating vP.

What about Passive and Reflexive Voice? These heads are not in complementary distribution, as reflexives can appear in passives:

(76) John was revealed to himself (by his analyst).

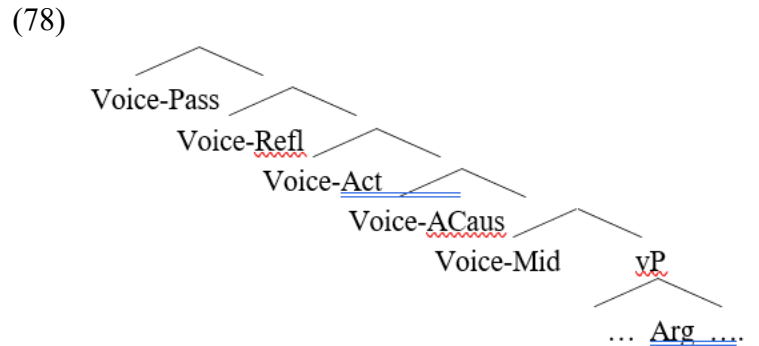
Furthermore, passives of ditransitives are possible (at least in some varieties of British English, including the second author’s):

(77) The books were given John.

Finally, it is clear that passives, as typically periphrastic constructions, interact with the auxiliary system, which we take to be located in the inflectional field. Hence we situate Voice-Passive as the highest head, with Voice-Reflexive in an intermediate position between this and Voice-Active; this may go some way towards explaining the observation made in the previous section that periphrastic passives are cross-linguistically very common, while other forms of periphrastic non-active voice are much rarer (although, as we have seen, they are found in Albanian-type languages). The features of Voice-Passive and Voice-Reflexive must be distinct, so that Voice-Reflexive’s feature does not prevent Voice-Passive from probing a vP-internal argument (which is realised as a *by*-phrase as in (76)). It is in this that Voice-Reflexive is a little different from the other Voice heads (recall that Voice-ACaus and Voice-Middle both constrain the argument structure of vP such that non-transitive vPs are impossible and so only

one argument can be probed, hence reflexives are impossible in anticausatives and middles).

So we arrive at the hierarchical structure for the putative “Voice field” as in (78):



We conclude our conjectures on the hierarchy of Voice heads here.

In the next section, we will try to extend our approach to the superficially very different voice systems found in Austronesian and Mandarin, adding also a brief note on Hebrew. In this way, we illustrate the typological potential of our approach.

9. A tentative extension to Austronesian and Mandarin voice systems

9.1. Austronesian

Looking first at Austronesian voice systems, these languages are very well-known for the complexity and richness of morphological marking of voice, and associated case and agreement alternations. This is particularly true of the Philippine languages. In (79), we illustrate a typical Philippine pattern from Puyuma. Example (79a) shows “Actor Voice” (AV); here the verb is marked by the infix *-em-*, the agent argument (‘the child’) is marked with *na*, traditionally known in the Austronesian literature as a “definite pivot” marker, and the direct object is marked accusative. The word order here as in all other examples is V>S>O>Oblique. In (79b-d), we see the third person nominative prefix *tu-*. In (79b) we have “Patient Voice”

(PV) marking in the form of the suffix *-aw* on the verb, the agent is marked nominative, while the patient receives the definite pivot marking *na*. In (79c), the verb is marked with the “Locative Voice” marking *-ay*, and the locative argument ‘road’ bears the definite pivot marking *na*. This argument also precedes the patient argument. Finally, in (79d), we have the “Circumstantial Voice” suffix *-anay*, and the instrumental argument has the definite pivot marking and precedes the patient.

(79) A typical Philippine pattern (Puyuma, Chen 2023):

- a. Selap na walak kana ramaraman i dalran dra saselap.
sweep<AV> DEF.PIVOT child DEF.ACC rubbish LOC road INDF.OBL broom
‘The child swept up the rubbish on the road with a broom.’ **(Actor Voice)**
- b. Tu=selap-aw kana walak na ramaraman i dalran dra saselap.
3.NOM=sweep-PV DEF.NOM child DEF.PIVOT rubbish LOC road INDF.OBL broom
‘The child swept up *the rubbish* on the road with a broom.’ **(Patient Voice)**
- c. Tu=selap-ay kana walak na dalran kana ramaramandra saselap
3.NOM=sweep-LV DEF.NOM child DEF.PIVOT road DEF.ACC rubbish INDF.OBL broom
‘The child swept up the rubbish on *the road* with a broom.’ **(Locative Voice)**
- d. Tu=selap-anay kana walak na saselap kana ramaraman i dalran.
3.NOM=sweep-CV DEF.NOM child DEF.PIVOT broom DEF.ACC rubbish LOC road
‘The child swept up the rubbish on the road with *the broom*.’ **(Circumstantial Voice)**

We propose the analysis of this system summarised in (80). Following Chen (2023), we take each voice to be associated with topic-marking of a different argument with the *na* definite pivot marker, which is in complementary distribution with the otherwise regular case-agreement marking of the relevant grammatical function. Thus, in Actor Voice, the external

argument is topic-marked, while the internal argument is accusative, and the obliques marked by the appropriate prepositions; this is schematised in (80a). In Patient Voice, (80b), the internal argument is topic-marked, the external argument is nominative, and the obliques are the same as in Actor Voice. Locative and Circumstantial Voice mark the locative and instrumental arguments, respectively, as topics, with the external argument nominative and the internal argument accusative.

- (80) a. Actor voice: EA-Topic IA-ACC
 b. Patient voice: IA-Topic EA-Nom
 c. Locative voice: LOC-Topic EA-Nom IA-Acc
 d. Circumstantial voice: INST-Topic EA-Nom IA-Acc

More specifically, we analyse the verb-initial order as verb-raising into the Clitic Voice field (in the sense of Sportiche 1996, see section 3); the morphological markers of the different voices are precisely exponents of the Sportichean Clitic Voice heads. The *na*-marked pivot/topic-arguments occupy SpecVoiceP with the Voice head agreeing with the Clitic Voice position, and thereby attracting the relevant argument to its Specifier. The external argument (when not a topic) precedes Voice, the internal argument (when not a topic) follows Voice.

The structures we assume for each voice seen in (79) and (80) are given in (81):²⁴

²⁴ Rackowski and Richards (2005) propose that voice alternations in Tagalog result from object shift and that voice morphology is Case agreement with the shifted object. Clearly this proposal is very similar in spirit to our proposals here, notably in that it makes Indo-European voice systems look more similar to the Austronesian voice system than has previously been thought.

- (81) a. [CIVoice [Cl-Voice V+Voice_{ACTOR}] [TP ... [VoiceP EA_{TOP} Voice_{ACTOR} [VP (EA) IA_{ACC}
LocP CircP]]]]
- b. [CIVoice [Cl-Voice V+Voice_{PATIENT}] [TP EA_{NOM} ... [VoiceP IA_{TOP} Voice_{PATIENT} [VP (EA) (IA)
LocP CircP]]]]
- c. [CIVoice [Cl-Voice V+Voice_{LOC}] [TP EA_{NOM} ... [VoiceP LocP_{TOP} Voice_{LOC} [VP (EA) IA_{ACC}
(LocP) CircP]]]]
- d. [CIVoice [Cl-Voice V+Voice_{CIRC}] [TP EA_{NOM} ... [VoiceP CircP_{TOP} Voice_{CIRC} [VP (EA) IA_{ACC}
LocP (CircP)]]]]

Here we see the Agree relation between the two Voice heads, namely, the high Clitic Voice head and the lower Voice head. The lower head is also endowed with an EPP feature which triggers movement to the Specifier of the (low) Voice head. Thus, strictly speaking, there are in fact two Agree relations, namely between the two Voice heads on the one hand, and the lower Voice head and the relevant argument on the other. Although we postulate that the EA raises to SpecTP when not the pivot/topic argument, we are not committed to the idea that SpecTP has the properties of an A-position in these languages.²⁵ Indeed, following Aldridge (2021), we consider that it is possible that the A- vs. A'-distinction is not operative in these languages (cf. also Van Urk 2015).

Now let us look back at the schemata in (68) and (69) that we gave for the Indo-European languages that we analysed in the preceding sections, from, as it were, “an Austronesian perspective”:

- (82) a. [VoiceP [Voice REFL] ... [VP V IA-self/body(part)]]
- b. [VoiceP [Voice A-CAUS] ... *from* P ... [VP V (IA)]]

²⁵ We thank Victoria Chen (personal communication) for drawing our attention to this point.

- c. [VoiceP [Voice MIDDLE] ... [AdvP ADV *for* P] ... [VP V (IA)]]
- d. [VoiceP [Voice PASSIVE] ... *by* P ... [VP V (IA)]]
- e. [VoiceP [Voice ACTIVE] ... [VP [VP V IA]]

The Case relation that we postulated as holding between Voice and the reflexive anaphor, or PPs, in (82a-d), and between Active Voice and the IA in (82e) clearly corresponds to the Agree relation just postulated between the lower Voice and the appropriate argument in (81). In these terms, we could consider the Anticausative in (82b) and the Passive in (82d) as two kinds of Patient voice.²⁶ The Middle in (82c) we could consider to be Experiencer Voice, somewhat analogous to the Austronesian Locative Voice (cf. Landau 2010 on the connection between Locatives and Experiencers, and the discussion in section 6). In the same vein, we could consider (82a) to be a special Reflexive Voice.²⁷

Despite these important similarities, there are two clear salient differences between Austronesian and Indo-European type voice systems. First, unlike in Indo-European, in Austronesian the verb raises to the high Clitic Voice head.²⁸ Second, in Austronesian, the low Voice head has an EPP feature which attracts the pivot/topic feature to its Specifier. Note that

²⁶ Victoria Chen (personal communication) reminds us that some Philippine-type languages have an Indo-European style active-passive alternation, which can co-occur with the Philippine-type alternation (see also Chen 2023). This intriguing observation is consistent with our observation that participial passives to some extent fall outside of the general Voice system we have described. Note also that at least in Romance SE-impersonals can co-occur with the participial passive: *Si è stati arrestati* (lit. ‘SI is been arrested’, i.e. ‘People were arrested.’). Whether these facts can be linked to our “cartographic” proposal in section 8, that Voice-Passive is the highest head in the “Voice field” (see (78)) remains to be seen.

²⁷ For details on reflexives in Austronesian languages, see Franssen (2010), who, crucial to our contention, notes a connection between verb-affixal reflexives and verb-affixal passives in these languages among other things.

²⁸ This also appears to be true in Hebrew, where we find non-active Voice realised as templatic morphology, as in (i), provided to us by an anonymous reviewer:

(i)	<i>nisraf</i>	<i>me-ha-šemeš</i>
	burn.UNACC	from-the-sun

Therefore, we treat templatic morphology as a form of exponent of the “high” Voice probe (see Shlonsky 1997 on verb movement to a high position in Hebrew). So, in our system, both clitic doubling and Voice doubling are instantiations of the Agree relation.

these differences represent very well-known forms of parametric variation, namely: verb-movement and EPP features.

An intriguing question concerns the role of the (high) Clitic Voices in the Indo-European languages. Both Albanian and Greek, and indeed also the Romance languages, show that the Clitic Voices can in fact be exponents of non-active Voice (see our discussion in the previous sections, in particular section 2). As for English, we have seen that Voice is consistently null, except in the participial passive, and so there is no bar to considering the Clitic Voice heads in English to be syntactically present but phonologically null. Therefore the Agree relations between the two Voice heads are the same across the Indo-European and the Austronesian languages.

9.2. Mandarin

Here we briefly consider how the Chinese *bei*-construction, often taken to be the analogue of the Indo-European passive, fits in the general picture that we have presented in this article. Huang (1999) and Huang, Li and Li (2009) propose that there are in fact two *bei*-constructions, namely: (i) the long passive, which is analogous to the English *tough*-construction, in which *bei* takes a CP-complement which features null-operator movement to SpecCP; and (ii) a short passive construction, which we take to have *bei* in Voice with a vP-complement. These two structures are shown in (83):

(83) a. Long passive:

Zhangsan_i bei [_{CP} OP_i [*Lisi da-le t_i*]]

Zhangsan BEI Lisi hit-PERF

‘Zhangsan was hit by Lisi.’

b. Short passive:

Zhangsan_i bei [_{VP} PRO_ida-le t_i]]

Zhangsan BEI hit-PERF

‘Zhangsan was hit.’

Here we leave aside the nature of the long passive (see Huang 1999 and Huang, Li and Li 2009 for detailed discussion). Concerning the short passive in (83b), we propose that *bei* corresponds to passive/Patient Voice. This raises the question of the status of PRO in (83b) in relation to the NP-movement relation that is standard in English and other languages in (short) passives. Following Chomsky et al. (2023), the distinction between “NP-trace” and “PRO” reduces to Form Copy in both cases, the difference being whether the higher copy (i.e., *Zhangsan* in (83b)) is internally or externally merged. Here we do not take a view on this question.

As a final point, it is worth noting that this account of the exponents of passive voice in Mandarin is consistent with the general “deep” analytic nature of Mandarin morphosyntax, as argued for in detail by Huang (2015).

10. Conclusions

We have argued that non-active Voices are licensors of abstract oblique Case, showing exponence/doubling patterns reminiscent of clitic doubling as analysed by Sportiche (1996); Baker, Johnson and Roberts’ (1989) analysis of passives anticipated this in the context of a rather different framework. As mentioned in the introduction, we view oblique Cases as interpretable features in contrast to structural Cases which are uninterpretable. The non-active Voices tend to show the same exponence/doubling pattern in a given language (with Albanian

a particularly interesting and complex case; Kallulli and Trommer 2011, Kallulli and Roberts, in press); we treat this as exponence of an oblique-licensing Voice. The nature of the Case-licensing affects the surface realisation of both the internal argument and, depending on precise assumptions about the position of the external argument, also the external argument. Furthermore, we have tentatively extended our approach beyond Indo-European to both Austronesian and Mandarin Chinese. In the former case, we have seen a striking deep similarity between Austronesian and Indo-European voice systems subject to relatively simple parametric variation involving verb-movement and EPP features. We have also seen that the Mandarin short *bei*-passive fits into our system, and is consistent with the deep analyticity of this language. We have also drawn attention to the similarities between Voice and AgrS and we sketched a possible cartographic analysis of the “Voice field”, in which each type of Voice represents a distinct head in a fixed structural hierarchy.

Two important further questions remain. First, what is the nature of implicit arguments in the various non-active constructions? Second, what is the status of impersonals? On the former, we discussed the nature of these arguments but we have left certain important questions open; see again Collins (2024) on implicit arguments in passives, Kallulli (2007) on implicit arguments in anticausatives, and Hoekstra and Roberts (1993) on implicit arguments in middles. On the latter, Blevins (2003) argues that impersonals are transitive structures; Roberts (2019) develops this idea and treats impersonals as transitives with an arbitrary external argument. However, this approach does not account for the fact that in many languages the exponents of impersonals tend to be syncretic to those for non-active Voice, e.g. Italian *si* (on which see Cinque 1988, D’Alessandro 2007).

Finally, we can point out that treating Voice alternation as oblique Case relations is consistent with Redundancy and Deficiency (see again Onea et al. 2023). A further consequence of this is that parametric variation in Voice systems, like that we have observed

between Austronesian and Indo-European, falls under the Redundancy-and-Deficiency-based approach to parameters developed in Kallulli and Roberts (2025b).

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